Evacuated Tube Solar Collector

Department of Mechanical and Energy Engineering

Author :Ekhlas Yousef Sead

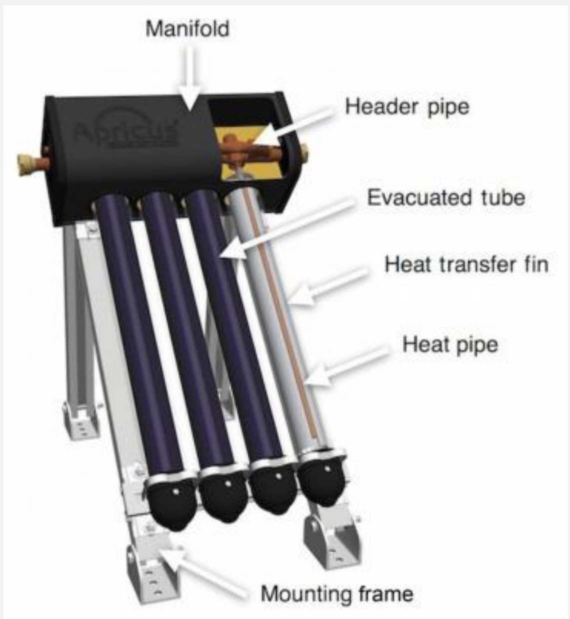
Under Supervision of : Dr. Giamal Mashina

Introduction

An **Evacuated Tube Solar Collector (ETSC)** is a high-efficiency solar thermal technology used to capture and convert solar energy into heat. It consists of parallel glass tubes with a vacuum layer that significantly reduces heat loss to the surrounding environment. This design allows the collector to perform effectively even in cold, cloudy, or windy conditions.
ETSC systems are widely used for **water heating, space heating, and industrial thermal applications** due to their superior thermal insulation, high absorption efficiency, and long operational lifespan. Compared to flat-plate collectors, evacuated tube collectors offer enhanced performance, especially in regions with variable climatic conditions, making them a sustainable and energy-efficient solution for renewable heat generation.

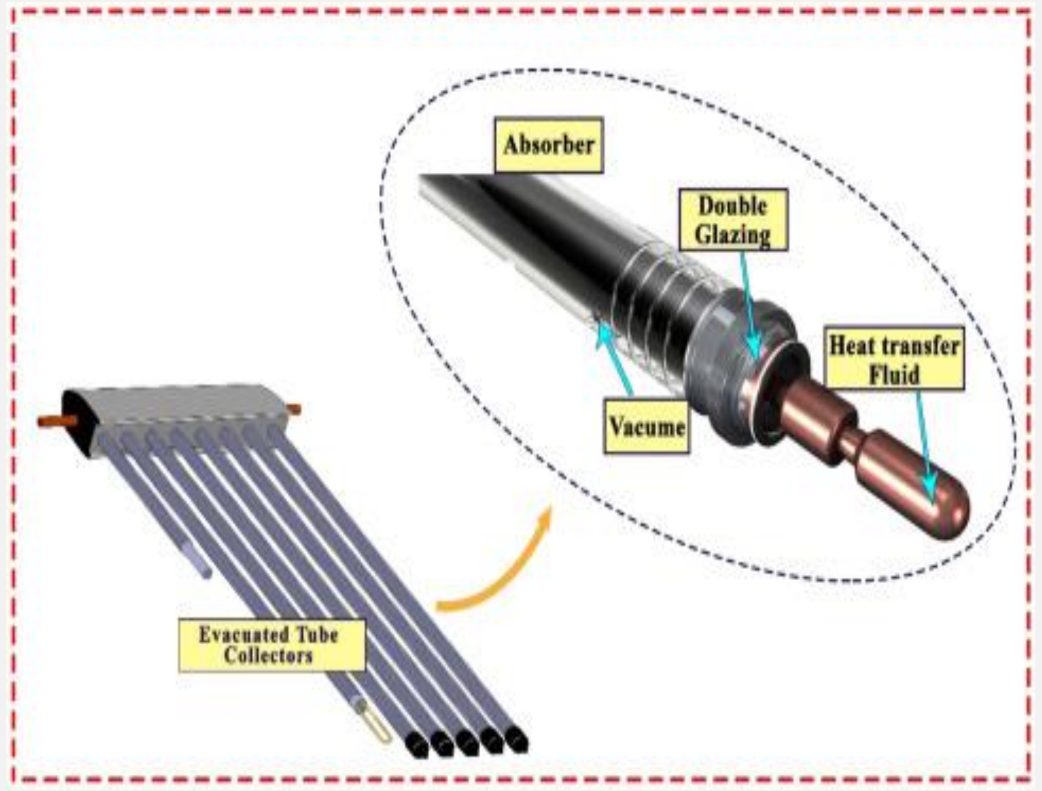
Components

Evacuated tube solar collector (ETSC), also known as Vacuum tube collectors, is a collector made up of evacuated glass tubes, aluminum fins, and a heat pipe. The selective coatings inside the evacuated tubes facilitate the absorption optimization; heat transfer in the tube to copper heat pipe is made possible by the fins. Heat is transferred to the water through the heat pipes. The system consists of a transparent glass tube arrange in rows parallel to each other and connected to a header pipe that permits circulation of working fluid and absorptions of generated heat by tubes (Fig. 5). As a result of small reflection on the tubes, solar radiation passes through the outside tube, and the inner tube absorbed heat. Heat loss is minimized by establishing a vacuum between the tubes fused on top of each other. Vacuum is an important factor that distinguishes ETSC from other collectors because it prevents heat loss and makes the collector a vast solar system for efficient absorption of solar energy

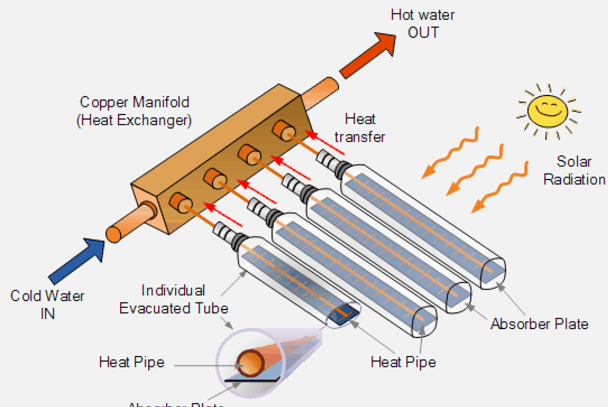


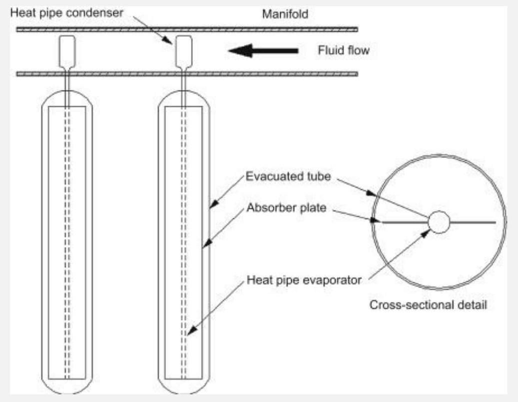
Evacuated Tube

These glass tubes are cylindrical in shape. Therefore, the angle of the sunlight is always perpendicular to the heat absorbing tubes which enables these collectors to perform well even when sunlight is low such as when it is early in the morning or late in the afternoon, or when shaded by clouds. Evacuated tube collectors are particularly useful in areas with cold, cloudy wintry weathers



how do solar evacuated tube collectors work?. Evacuated tube collectors are made up of a single or multiple rows of parallel, transparent glass tubes supported on a frame. Each individual tube varies in diameter from between 1" (25mm) to 3" (75mm) and between 5′ (1500mm) to 8′ (2400mm) in length depending upon the manufacturer.





each tube consists of a thick glass outer tube and a thinner glass inner tube, (called a "twin-glass tube") or a "thermos-flask tube" which is covered with a special coating that absorbs solar energy but inhibits heat loss. The tubes are made of borosilicate or soda lime glass, which is strong, resistant to high temperatures and has a high transmittance for solar irradiation.

Unlike flat panel collectors, evacuated tube collectors do not heat the water directly within the tubes. Instead, air is removed or evacuated from the space between the two tubes, forming a vacuum (hence the name evacuated tubes).

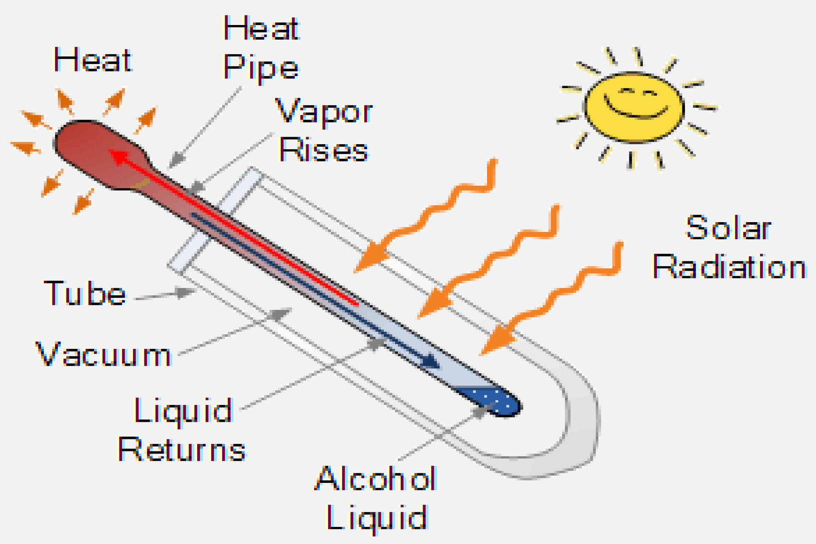
This vacuum acts as an insulator reducing any heat loss significantly to the surrounding atmosphere either through convection or radiation making the collector much more efficient than the internal insulating that flat plate collectors have to offer. With the assistance of this vacuum, evacuated tube collectors generally produce higher fluid temperatures than they're flat plate counterparts so may become very hot in summer.

Heat Pipe Evacuated Tube Collector

In heat pipe evacuated tube collectors, a sealed heat pipe, usually made of copper to increase the collectors efficiency in cold temperatures, is attached to a heat absorbing reflector plate within the vacuum sealed tube.

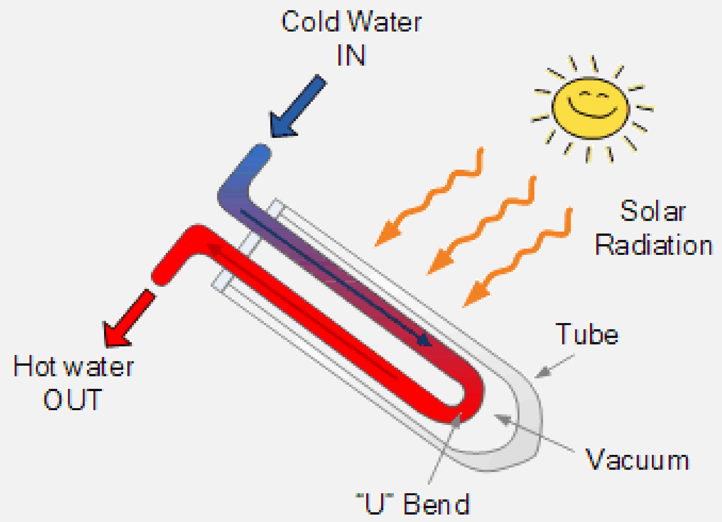
The hollow copper solar heat exchanger design within the tube is evacuated of air but contains a small quantity of a ˏow pressure alcohol/water liquid plus some additional additives to prevent corrosion or oxidation.
heat pipe collector

This vacuum enables the liquid to vaporise at very lower temperatures than it would normally at atmospheric pressure. When sunlight in the form of solar radiation hits the surface of the absorber plate inside the tube, the liquid in the heat pipe quickly turns into a hot vapour type gas due to presence of the vacuum. As this gas vapour is now lighter, it rises up to the top portion of the pipe heating it up to a very high temperature.



The top part of the heat pipe, and therefore the evacuated tube is connected to a copper heat exchanger called the "manifold". When the hot vapours still inside the sealed heat tube enters the manifold, the heat energy of the vapour is transferred to the water or glycol fluid flowing through the connecting manifold. As the hot vapour looses energy and cools, it condenses back from a gas to a liquid flowing back down the heat pipe to be reheated.

The heat pipe and therefore the evacuated tube collectors must be mounted in such a way as to have a minimum tilt angle (around 30o) in order for the internal liquid of the heat pipe to return back down to the hot absorber plate at the bottom of the tube. This process of converting a liquid into a gas and back into a liquid again continues inside the sealed heat pipe as long as the sun shines.



Benefits

Although the cost of installation of ETSC is very high, they offer many benefits that outweigh the investments and are worth considering for domestic and commercial applications. Those benefits include wider collectors angle and more absorbing time, high heat retention on the tubes, high tolerance for strong winds, and endurance for freezing conditions. ETSC is one of the most expensive solar collectors. Still, they are more reliable, efficient, and cost-effective, giving their performance in extremely cold conditions where other collectors suffer a decline in performance due to anti-freezing. Owing to these characteristics, ETSC is very useful in cloudy wintry weathers such as Canada and the Northern part of the United States.

- High thermal efficiency due to vacuum insulation
- Minimal heat loss in cold or windy conditions
- Performs well in low sunlight and cloudy weather
- Long service life and durable glass construction
- Modular design allows easy replacement of damaged tubes
- Requires less roof space compared to flat-plate collectors
- Environmentally friendly and reduces energy costs

Conclusion

Evacuated tube solar collectors represent a highly efficient and advanced technology for solar thermal energy utilization. By incorporating vacuum-insulated glass tubes, these systems significantly reduce heat losses and maintain high thermal performance even in cold, windy, or low-sunlight conditions. This makes them particularly suitable for regions with variable climates and for applications requiring consistent heat output.

In addition to their superior efficiency, evacuated tube collectors are durable, modular, and easy to maintain, allowing for long-term operation with minimal performance degradation. Their ability to deliver reliable and environmentally friendly thermal energy contributes to reduced dependence on conventional energy sources, lower operating costs, and decreased carbon emissions. Overall, evacuated tube solar collectors provide a sustainable and practical solution for meeting modern energy demands while supporting the global transition toward renewable energy technologies.

References

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